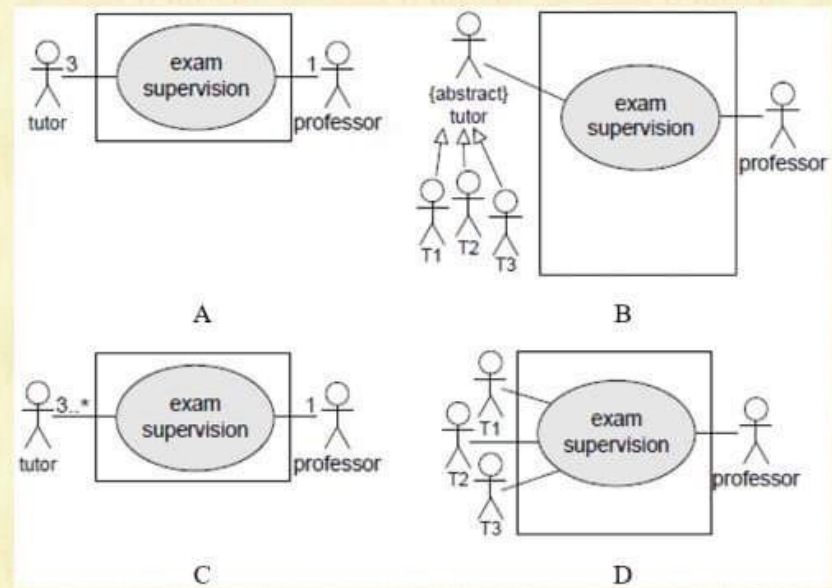


Time Remaining
0:30:42

9
Marks: 1
How do you model the following situation with a UML 2 use case diagram:
An exam supervision is done by one professor and three tutors.



Choose one answer.

- ☒ a. D
- ☐ b. A
- ☐ c. C
- ☐ d. B

10
Marks: 1
Which GRASP pattern do you use to decide who is going to handle the incoming system events?

Choose one answer.

- ☐ a. Adapter
- ☐ b. Controller
- ☐ c. Information Expert
- ☐ d. Low coupling

THI KẾT THÚC HỌC PHẦN OOAD - HỌC KỲ II, NĂM HỌC 2017-2018

Time Remaining
0:32:50

1

Which of the following statements are true?

Marks: 1

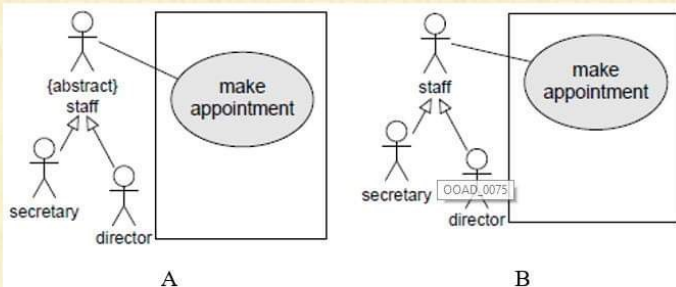
Choose one answer.

- ☐ a. According to Rebecca Wirfs-Brock, a stereotype is a high-level classification of an object such as "controller" and "coordinator".
- ☐ b. All of the mentioned.
- ☐ c. Within UML, stereotypes are used for meta-model extensions.
- ☒ d. According to Jacobson, class stereotypes are: interface object, control object and entity object.

2

How do you model the following situation with a UML use case diagram:
The directors or his secretary are allowed to make an appointment, but nobody else.

Marks: 1

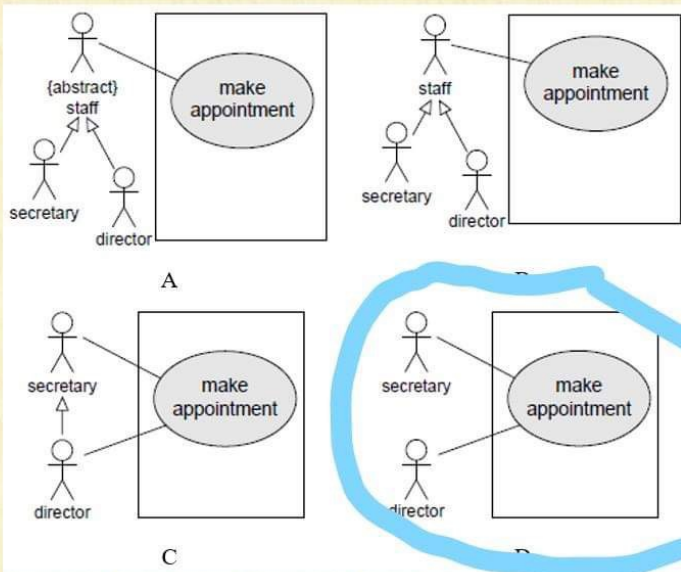


2

Marks: 1

How do you model the following situation with a UML use case diagram:
The directors or his secretary are allowed to make an appointment, but nobody else.

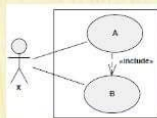
Time Remaining
0:32:00



19

Marks: 1

The following <<include>>-relationship means, that ...



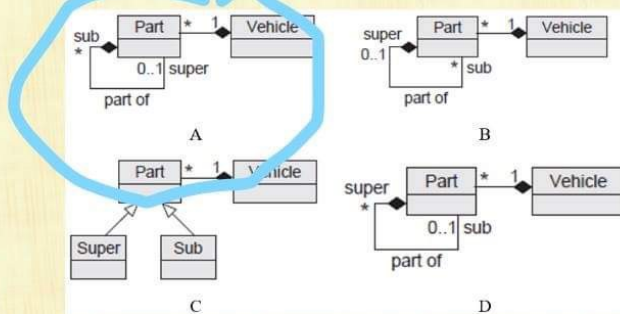
Choose one answer.

- ☐ a. A cannot be executed without B
- ☒ b. B cannot be executed without A
- ☐ c. B can be executed instead of A
- ☐ d. A might or might not invoke use case B

20

Marks: 1

How do you model the following situation with a UML class diagram: A vehicle comprises of multiple parts. Each part can be built in a maximum of one super-part. One super-part comprises of multiple sub-parts.



Choose one answer.

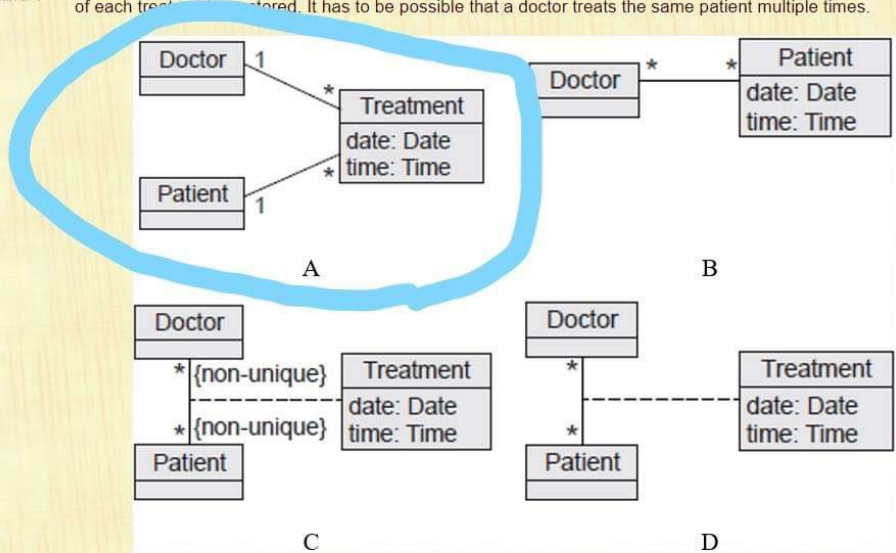
- ☐ a. D
- ☐ b. B

5

Marks: 1

How do you model the following situation with a UML class diagram: A doctor treats multiple patients, a patient can be treated by multiple doctors. To document which doctor has when treated which patient, the date and time of each treatment is stored. It has to be possible that a doctor treats the same patient multiple times.

Time Remaining
0:31:25



Choose one

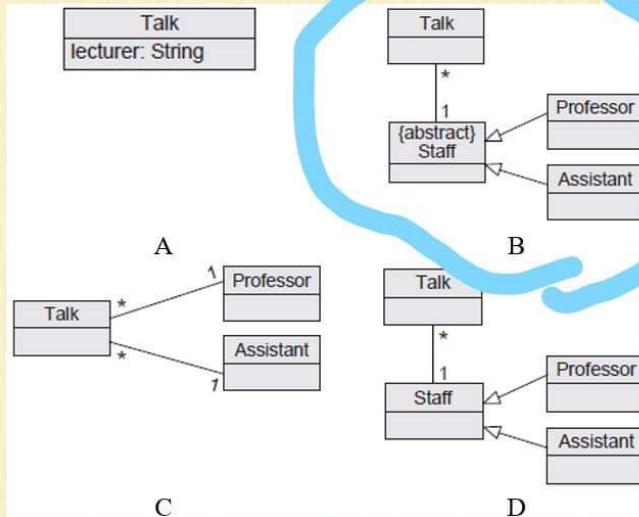
☐ a. D

3

Marks: 1

How do you model the following situation with a UML class diagram? A talk is delivered by a professor or by an assistant.

Time Remaining
0:31:37



Choose one
answer

☐ a. D

4 Which of the following statement is false?

Marks: 1

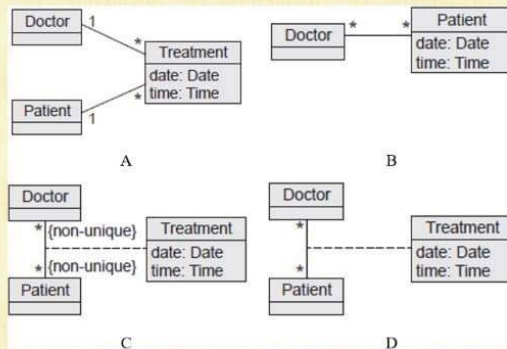
Choose one answer.

- ☐ a. When the whole is deleted, parts in aggregation are also removed.
- ☐ b. All of the mentioned.
- ☐ c. Both Aggregation and Composition are 'part-of' relationship.
- ☐ d. Aggregation is a special kind of Association.

Time Remaining
0:30:54

5 How do you model the following situation with a UML class diagram: A doctor treats multiple patients, a patient can be treated by multiple doctors. To document which doctor has when treated which patient, the date and time of each treatment are stored. It has to be possible that a doctor treats the same patient multiple times.

Marks: 1

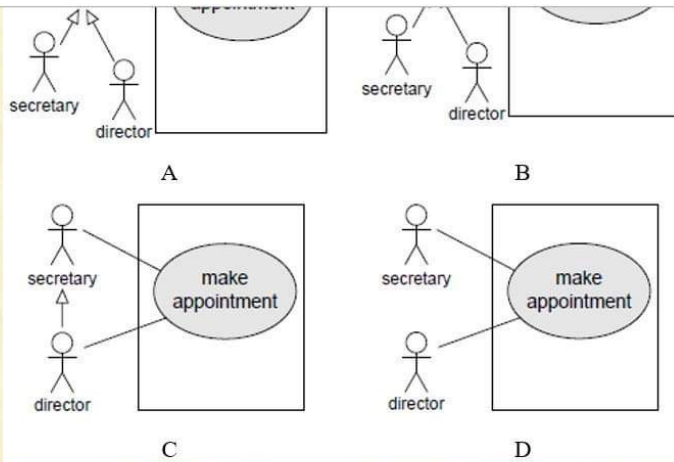


Choose one answer.

- ☐ a. D
- ☐ b. B
- ☒ c. A
- ☐ d. C

6 You are given the following sequence diagram. Which traces are possible?

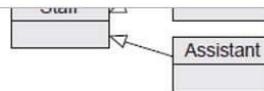
Time Remaining
0:31:47



Choose one
answer.

- ☐ a. A
- ☐ b. C
- ☒ c. D
- ☐ d. B

Time Remaining
0:31:29



C

D

Choose one answer.

- ☐ a. D
- ☐ b. C
- ☐ c. A
- ☒ d. B

4

Which of the following statement is false?

Marks: 1

Choose one answer.

- ☐ a. When the whole is deleted, parts in aggregation are also removed.
- ☐ b. All of the mentioned.
- ☐ c. Both Aggregation and Composition are 'part-of' relationship.
- ☐ d. Aggregation is a special kind of Association.

5

How do you model the following situation with a UML class diagram: A doctor treats multiple patients, a patient can be treated by multiple doctors. To document which doctor has when treated which patient, the date and time of each treatment are stored. It has to be possible that a doctor treats the same patient multiple times.

Marks: 1



23

Marks: 1

What do you think attributes describe?

- a) The state of the object
- b) The properties of the object
- c) The function of the object
- d) The characteristics of the object

Choose one answer.

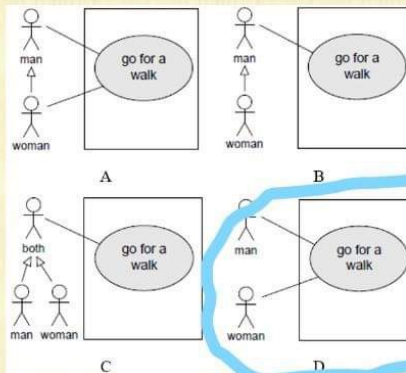
- ☐ a. a, b and d
- ☒ b. a, b and d
- ☐ c. a and c
- ☐ d. a, b and c

Time Remaining
0:29:53

24

Marks: 1

How do you model the following situation with a UML use case diagram:
A man and a woman go for a walk



13

Marks: 1

Which statements are false for a use case?

Choose one answer.

- ☐ a. a use case can be traced to a discrete goal
- ☒ b. a use case captures some user-visible and non-visible functions
- ☐ c. good source for identifying use cases is external events
- ☐ d. a use case may have many scenarios

Time Remaining
0:30:31

14

Marks: 1

Which of the following statements are true for Activity diagrams?

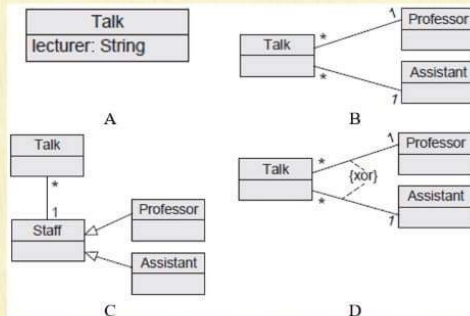
Choose one answer.

- ☐ a. can be used to explore/discover parallel activities
- ☐ b. do not tell you who does what and are difficult to trace back to object models
- ☐ c. all of the mentioned
- ☐ d. can be used to depict workflow for a particular business activity

15

Marks: 1

How do you model the following situation with a UML class diagram: Each talk is done by a professor or by an assistant.



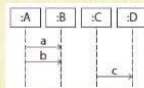
Choose one answer.

- ☐ a. D

Time Remaining
0:30:48

6 You are given the following sequence diagram. Which traces are possible?

Marks: 1



Choose one answer.

- ☐ a. $a \rightarrow c \rightarrow b$
- ☐ b. $b \rightarrow c \rightarrow a$
- ☐ c. $c \rightarrow b \rightarrow a$
- ☐ d. $b \rightarrow a \rightarrow c$
- ☐ e. Another Answer

7 Use case description contents includes ?

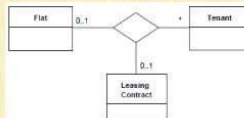
Marks: 1

Choose one answer.

- ☐ a. Use case name and number
- ☒ b. All of the mentioned
- ☐ c. Actors
- ☐ d. Stakeholder and needs

8 You are given the following clipping of a UML class diagram. Which of the following statements are true?

Marks: 1



Choose at least one answer.

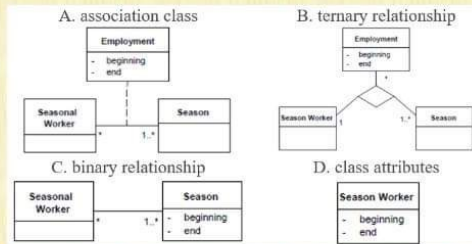
- ☒ a. One flat can be rented by multiple tenants with different leasing contracts.
- ☒ b. One flat can be rented by multiple tenants with the same leasing contract.
- ☐ c. One tenant can rent two flats with one leasing contract.
- ☐ d. One tenant can rent multiple flats with different leasing contracts.

Time Remaining
0:30:36

11

Marks: 1

How do you model the following situation with a UML class diagram: A season worker may be employed in one or several seasons, the beginning and the end of each employment period is saved.



Choose one answer.

- ☒ a. A
- ☐ b. C
- ☐ c. D
- ☐ d. B

12

Marks: 1

Which of the following statements about the given diagram clipping are true?



Choose at least one answer.

- ☒ a. B inherits all of A's associations.
- ☐ b. A inherits all of B's associations.
- ☐ c. A can execute the same use cases as B.
- ☒ d. B can execute the same use cases as A.

13

Which statements are false for a use case?

21

Marks: 1

Which of the following statements are true?

Choose one answer.

- ☐ a. None of the above.
- ☐ b. As models are refined, it is acceptable to use the same set of class diagrams for different perspectives: conceptual, specification and implementation.
- ☒ c. Maintaining different models for different perspectives will run into a risk of inconsistency among models.
- ☐ d. All of the above.

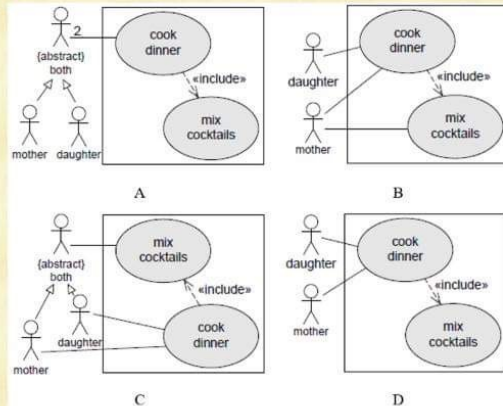
Time Remaining
0:30:05

22

Marks: 1

How do you model the following situation with a UML use case diagram:

A mother cooks dinner together with her daughter. In the course of that, the mother also always has to mix the cocktails.



Choose one answer.

- ☒ a. B
- ☐ b. C
- ☐ c. D

23

Marks: 1

What do you think attributes describe?

- a) The state of the object
- b) The properties of the object
- c) The function of the object
- d) The characteristics of the object

Choose one answer.

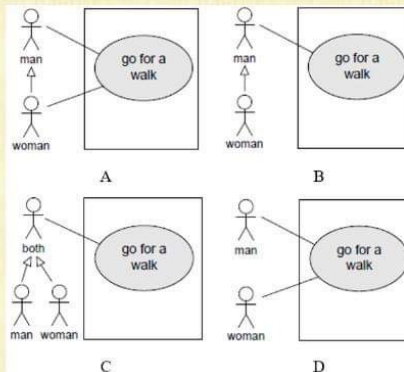
- ☐ a. a and d
- ☒ b. a, b and d
- ☐ c. a and c
- ☐ d. a, b and c

Time Remaining
0:29:53

24

Marks: 1

How do you model the following situation with a UML use case diagram:
A man and a woman go for a walk



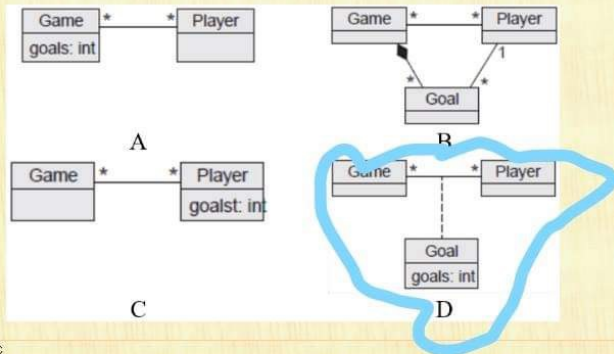
d. Name

Time Remaining
0:29:40

29

Marks: 1

How do you model the following situation with a UML class diagram: During one soccer season, multiple players participate in multiple games. Each player scores in each game a certain number of goals.



Choose one answer.

- ☐ a. C
- ☐ b. A
- ☐ c. B
- ☐ d. D

30

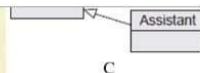
Marks: 1

Which of the following statements about the given diagram are true?



Choose one answer.

- ☐ a. If an instance of B is deleted, all contained instances of A are also deleted.
- ☒ b. If an instance of A is deleted, all contained instances of B are also deleted.
- ☐ c. A is part of B.



Time Remaining
0:30:22

Choose one answer.

- ☐ a. D
- ☐ b. B
- ☐ c. A
- ☒ d. C

16

Marks: 1

In order to model the relationship "a hotel has rooms", between Hotel and Room, you could use:

Choose one answer.

- ☐ a. Association
- ☐ b. Composition
- ☒ c. Generalization
- ☐ d. Aggregation

17

Marks: 1

Which statement is true for an Actor?

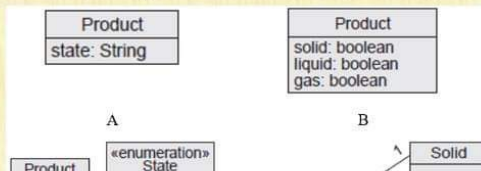
Choose one answer.

- ☐ a. An actor cannot be inherited.
- ☐ b. An actor is a person interacts with the system.
- ☐ c. An actor is a person plays a role in a use case.
- ☒ d. An actor is a role a user plays with respect to the system.

18

Marks: 1

How do you model the following situation with a UML class diagram: Every product is in one of the three following states: solid, liquid or gas.



Time Remaining
0:29:48

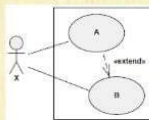
Choose one answer.

- ☐ a. B
- ☐ b. D
- ☐ c. A
- ☐ d. C

25

Marks: 1

The following <<extend>>-relationship means, that ...



Choose one answer.

- ☐ a. can extend A....
- ☐ b. can be executed instead of A....
- ☐ c. always has to invoke A.
- ☒ d. B might or might not invoke A....

26

Marks: 1

If you want to plan project activities such as developing new functionalities or test cases, which of the following OOAD artifacts is the most useful?

Choose one answer.

- ☐ a. Package diagrams.
- ☒ b. Sequence diagrams.
- ☐ c. Use cases.
- ☐ d. Domain model.

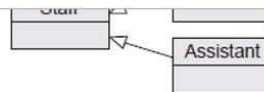
27

Marks: 1

How do you model the following situation with a UML use case diagram: A person goes for lunch. In the course of that it might be necessary that the person withdraws cash from an ATM.



Time Remaining
0:31:29



C

D

Choose one answer.

- ☐ a. D
- ☐ b. C
- ☐ c. A
- ☐ d. B

4

Which of the following statement is false?

Marks: 1

Choose one answer.

- ☒ a. When the whole is deleted, parts in aggregation are also removed.
- ☐ b. All of the mentioned.
- ☐ c. Both Aggregation and Composition are 'part-of' relationship.
- ☐ d. Aggregation is a special kind of Association.

5

How do you model the following situation with a UML class diagram: A doctor treats multiple patients, a patient can be treated by multiple doctors. To document which doctor has when treated which patient, the date and time of each treatment are stored. It has to be possible that a doctor treats the same patient multiple times.

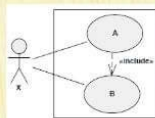
Marks: 1



19

Marks: 1

The following <<include>>-relationship means, that ...



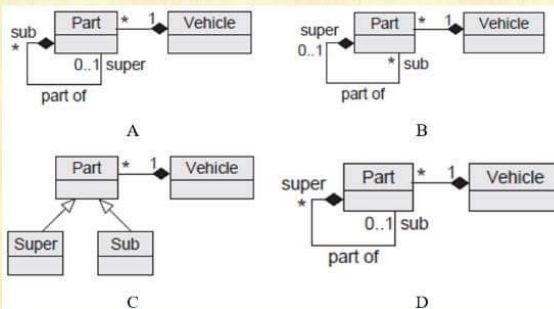
Choose one answer.

- ☐ a. A cannot be executed without B
- ☒ b. B cannot be executed without A
- ☐ c. B can be executed instead of A
- ☐ d. A might or might not invoke use case B

20

Marks: 1

How do you model the following situation with a UML class diagram: A vehicle comprises of multiple parts. Each part can be built in a maximum of one super-part. One super-part comprises of multiple sub-parts.



Choose one answer.

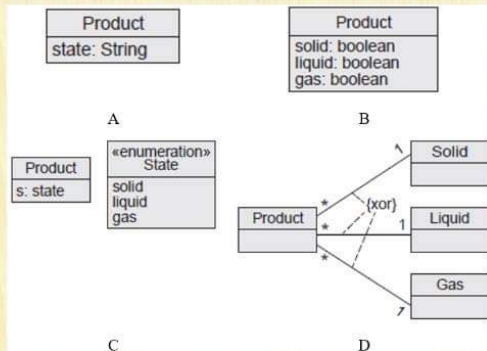
- ☐ a. D
- ☐ b. B

18

Marks: 1

How do you model the following situation with a UML class diagram: Every product is in one of the three following states: solid, liquid or gas.

Time Remaining
0:30:18



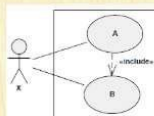
Choose one answer.

- ☐ a. B
- ☒ b. C
- ☐ c. A
- ☐ d. D

19

Marks: 1

The following <<include>>-relationship means, that ...

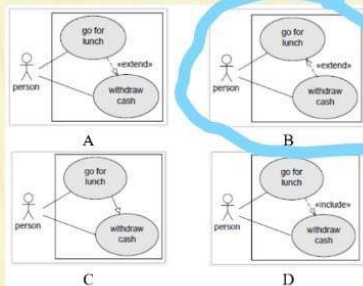


27

Marks: 1

How do you model the following situation with a UML use case diagram: A person goes for lunch. In the course of that it might be necessary that the person withdraws cash from an ATM.

Time Remaining
0:29:43



Choose one answer.

- ☐ a. C
- ☐ b. D
- ☐ c. A
- ☐ d. B

28

Marks: 1

Which of the following optional information can not be given on an association between classes?

Choose one answer.

- ☒ a. A small filled triangle near the association name to show in which way to "read" the name.
- ☐ b. Multiplicity at the end of an association.
- ☐ c. An arrow at the end to show navigability.
- ☐ d. Name

29

Marks: 1

How do you model the following situation with a UML class diagram: During one soccer season, multiple players participate in multiple games. Each player scores in each game a certain number of goals.

